Trade-Offs and Synergies for Agriculture and Environmental Outcomes in the Tropics

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Introduction

Global population is currently 8 billion, projected to increase by 34 percent by 2050 and to plateau at 10.9 billion by 2100 (UN 2022). Feeding a growing population presents numerous challenges, including the possible need to expand land under agriculture at the expense of tropical forest and to increase the intensity of agricultural inputs such as water and fertilizer. Tropical areas include many lower-income countries that are highly dependent on agriculture. Expanding farm area could generate water-contaminating runoff, CO_2 and methane emissions, and the loss of biodiversity and soil carbon. A number of challenges affect this trade-off between agriculture and the environment. Climate change makes future food supply more uncertain. Global trade has trended toward greater integration, fundamentally changing where food is produced. Enhanced agricultural technologies are changing the geography and environmental costs of food production.

Given these complexities, the relationship between increased agricultural production and environmental degradation is a central challenge to human well-being. Will feeding the world leave it with less forest, contaminated air and water, and decreased biodiversity? Are there pathways to food security that also conserve or improve environmental quality? This article compiles recent evidence on the relationship between agriculture and the environment, focusing on the tropics, where potential trade-offs between food production and environmental quality are particularly acute because much larger shares of the population

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depend upon agriculture for work and subsistence. Although land use is our primary metric of environmental impact, we also touch on associated ecosystem services (or environmental services) such as soil and air quality. Land-use change features prominently because the conversion of forest to crops represents an exchange of agricultural production for such environmental services. In addition, with the spread of large-scale satellite-based metrics of forest, deforestation is one of the most frequently measured environmental outcomes in the literature.

Recent remote sensing estimates show that forest area decreased by 2.4 percent between 2000 and 2020 (a loss of more than 1 million km^2) and that agricultural land area increased by 11.5 percent (an area of 1.25 million km^2; Potapov et al. 2022). Figure 1 shows the evolution of agricultural land use over time for tropical and nontropical countries, using administrative data from the Food and Agriculture Organization and remotely sensed measures of deforestation from Global Forest Watch.[1] Although agricultural land shares in nontropical countries have declined over the past half-century, tropical agricultural land shares have climbed. Forest cover has decreased in these same countries, remaining constant in nontropical areas.

Estimates from 2015 indicate that forests are mostly replaced by cropland and pasture (Tyukavina et al. 2015). The simple comparison between agricultural and forest area suggests that increases in the former come at the expense of the latter in many parts of the world, with the highest areas of net forest loss in South America (440,000 km^2) and Africa (320,000 km^2) and the largest areas of agricultural expansion in Africa (470,000 km^2), followed by South America (340,000 km^2; Potapov et al. 2022).

Our central contribution is to provide a deep dive into food production as a cause of land-use change in low- and middle-income countries, highlighting the recent rapid expansion of causal evidence in this space. Our work complements a number of existing reviews that focus broadly on the causes and consequences of land-use change around the world. Barbier (2001) summarizes the economics of deforestation and land use as of 2001. Jayachandran (2022) examines how economic development influences the environment, dedicating some space to agricultural productivity as a channel for environmental effects. Villoria, Byerlee, and Stevenson (2014) examine the effects of agricultural technology on deforestation, and García (2020) provides a report focused on the environmental effects of agricultural intensification—the production of more food per hectare (ha) of farmland through various techniques such as irrigation or the increased use of chemical inputs. Most recently, Balboni et al. (2023) present a broad review of economic theory and empirical analysis associated with tropical deforestation. Our work covers some aspects of land-use change associated with agriculture but does not highlight environmental effects of the broad suite of policies intended to increase food production. Finally, Busch and Ferretti-Gallon (2017, 2023) conduct a meta-analysis (synthesizing results from multiple studies) of drivers of land-use change.

By contrast, our approach is a narrative review (Greenhalgh, Thorne, and Malterud 2018). We provide summary along with interpretation and critique, including assessments of the internal and external validity of existing studies, and we frame our analysis with economic theory. Our analysis complements research that focuses on the drivers of land-use change. Such

[1] All figures use FAOStat (2024) data, except where noted. For each data series, countries with more than 5 years of missing data were removed. Countries are classified as tropical or nontropical, as in appendix A (available online). We note final sample sizes in the caption of each figure.

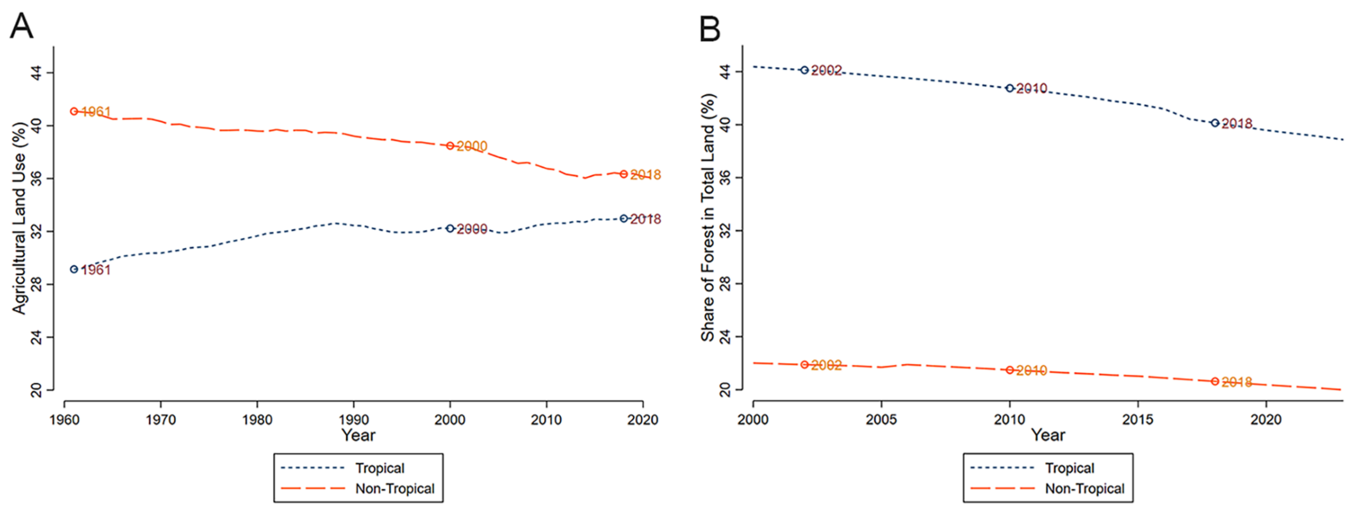

Figure 1 Evolution of agricultural and forest land use. Panel *A* illustrates the evolution of agricultural land use as a percentage of each country's total land area, from 1961 to 2021. Sample: 158 countries (62 nontropical, 96 tropical). Panel *B* shows the evolution of the forest share as a percentage of each country's total land area from 2000 to 2023 and includes data from Global Forest Watch (2024). The forest share for each year and country was calculated using the baseline forest area in hectares in 2000, adjusted annually by subtracting deforestation in hectares. Sample: 180 countries (94 tropical, 86 nontropical).

research demonstrates, for example, that deforestation is more likely when land has less slope and greater fertility (e.g., Busch and Ferretti-Gallon 2017). Although this can be helpful in targeting policies toward high-risk areas, it does not clarify the trade-offs inherent in the design of such policies, nor does it assess whether interventions also improved food production.

The next section presents a basic theoretical model and contextual descriptive evidence, which we then link to recent micro-based findings. The subsequent section shifts to the global context, focusing on macroeconomic and trade-related evidence. Finally, we conclude by cataloging interventions that generate positive agricultural production and environmental outcomes, highlighting gaps in the literature, and outlining key areas for future research based on our review.

Organizing Framework

Our discussion begins with the classical Von Thünen model of land rent, in which land is allocated to the use with the highest rent. Rent, in this context, reflects the value of a piece of land in a specific location at a specific point in time. It is equal to the revenues earned from a specific land use minus the costs of inputs and the cost of transporting the output to the market. This is a standard approach in land economics, and it has frequently been used to give context to the feedback between deforestation-reducing policies and agricultural production (Angelsen 2010). In this article, we consider the opposite dynamic: how agricultural output can be increased while conserving or restoring natural landscapes. The land-rent model can accommodate a number of land uses, but to fix ideas, we will use three: intensive agriculture (producing more food per hectare), extensive agriculture (expanding the area under cultivation), and forests. Further, this model is preferable to some other traditional models of forest loss (see Balboni et al. 2023) because the purpose of most tropical deforestation is not to harvest trees but rather to plant new crops, graze animals, and expand urban boundaries (Jayathilake et al. 2021).

Figure 2 shows the simplest schematic of these relationships. This figure shows the relationship between land rent and distance to the center of the nearest city. Each of the rent curves (shown as straight lines) is downward sloping, meaning that rent is higher for shorter distances. This follows from the assumption that there are nonzero costs of transporting goods to the market at the city center (the graph's origin). The hierarchy of land uses has the highest rents coming from intensive agriculture (R^{ai}) up to point A^*, then extensive agriculture (R^{ae}) up to F, and finally forest (R^f). The forest frontier—not in a geographic sense, but as the point where a landowner is indifferent between possible land uses—exists at the point at which extensive agriculture and forest yield the same level of rent. Private forest rents include forest use value (timber, local ecosystem services such as shade, and nontimber forest products such as edible plants) and future development value. The model easily extends to include broader forest ecosystem values as an externality—that is, a benefit or cost that is not directly valued by the decision-making landholder.

Increases in agricultural production result from changes in output or input prices, technology, or institutional factors affecting production. Output prices are prices for the commodities that farmers produce. Inputs include water, seeds, labor, fertilizer, pesticides, and the transportation of goods to market and of inputs to the farm. Relevant prices include

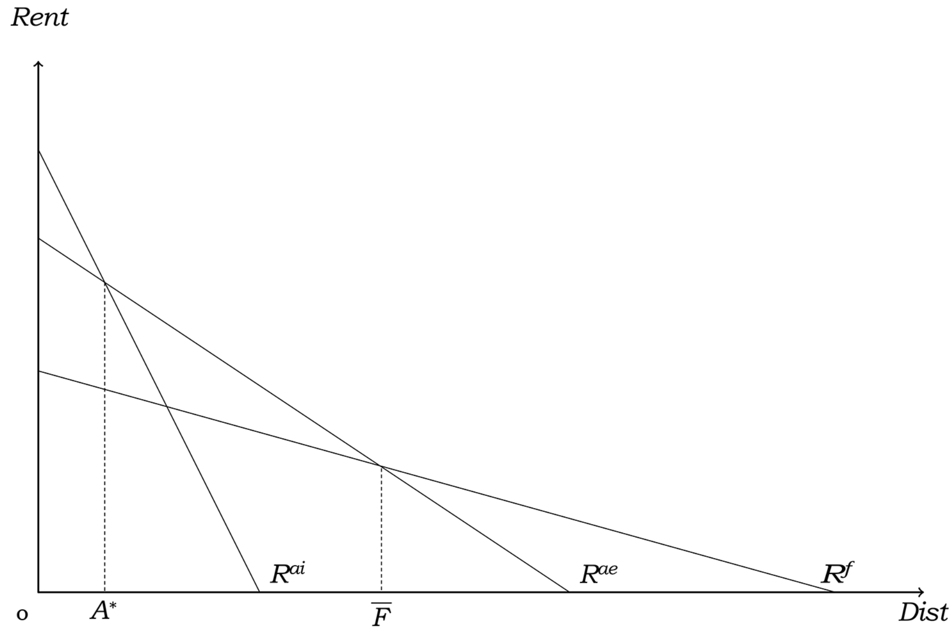

Figure 2 Simple land-rent model. The *Y*-axis measures rent in various land uses: intensive agriculture (*ai*), extensive agriculture (*ae*), and forest (*f*). The *X*-axis is distance to market.

wages, as well as the interest rate and other costs of obtaining credit to buy inputs. Transport costs may rise or fall. Technological change may allow for the production of more output using the same amount of inputs—this is effectively a decrease in the cost of production overall. Further, institutional changes that improve certainty around future agricultural rents (e.g., improved land tenure rights, which are relevant in developing-country contexts where land rights are often customary rather than formal) may effectively increase rent. Labor and credit are often affected by frictions, such as laborers not knowing about a job opportunity or farmers who do not have secure property rights to serve as collateral to obtain credit. In the simplest version of this model, where there are no frictions in labor or credit markets, we highlight a prediction of the model that is particularly relevant to the tropics.

To begin, take the case of former colonies situated in tropical regions, where colonizing countries were concerned about local land occupation and border control. Colonial regimes enacted policies that aimed to settle the population in the interior regions of these countries. Importantly, resettlement incentives primarily encouraged extensive agriculture. In our framework, this can be understood as having a much flatter extensive agricultural (R^{ae}) rent gradient than that of intensive agriculture (R^{ai}); that is, distance to market has a greater effect on intensive than extensive agriculture. As a result, population growth and the expansion of new cities in the country's interior cleared land for extensive agriculture (Pichon 1997; Barbier 2004).

Now, suppose a scenario where the Green Revolution, marked by the introduction of new technology and the advancement of intensive agriculture, encounters a substantial amount of preexisting cleared land. The curve R^{ai} flattens significantly, increasing the value of intensive agriculture farther out from the origin. Meanwhile, agricultural extensification was already profitable such that, even under the flatter R^{ai} curve, both intensive and extensive agriculture were taking place closer to the market than was forest production ($A^* < F$). In this case, innovation has the potential to induce more efficient land use, resulting in improved production through yield gains without the need to expand deforestation. Historical patterns of extensification create low-productivity agricultural zones that can act as a buffer to absorb continuous improvements in R^{ai} without altering the extent of forests.

On the other hand, in areas where extensive agriculture is not prevalent, improvements in the prospects of intensive agriculture can lead to a distinct outcome. In these areas, yield gains increase agricultural profitability compared with forested areas and can thus lead to additional deforestation. This phenomenon is commonly known as the "Jevons paradox." The role of history is therefore pivotal in understanding the connection between agricultural productivity and deforestation, and whether the phenomenon of the Jevons paradox emerges (Phalan et al. 2016).

From figure 1, we know the area in agriculture has been decreasing in nontropical countries and increasing in tropical countries. Figure 3 shows that land productivity, measured in agricultural output per hectare, increased more quickly for nontropical versus tropical countries until 2000. After 2000, nontropical agricultural productivity flattens, and tropical productivity inflects upward.

Taken together, these figures suggest that growth in nontropical, intensive agricultural technologies enabled smaller land footprints with greater productivity between 1960 and

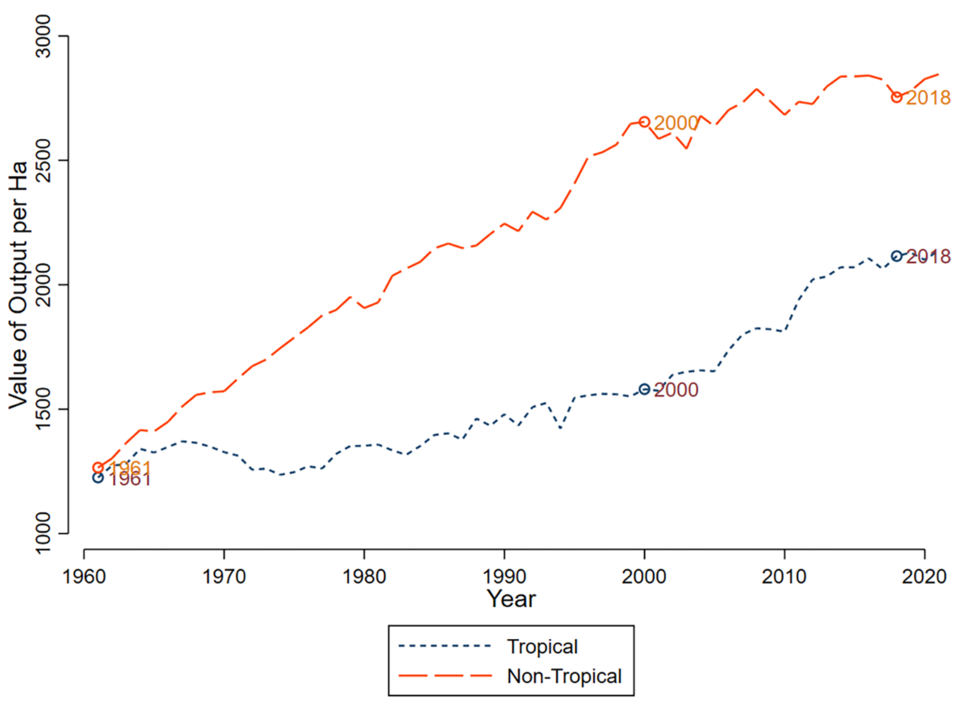

Figure 3 Evolution of output value per hectare of agricultural land. Output value is measured by gross production value in constant 2014–2016 thousand US dollars; agricultural land is measured in thousands of hectares. Output value per hectare is expressed in constant USD/ha. Sample: 115 countries (48 nontropical, 67 tropical).

2000. These productivity gains seem not to have led to a Jevons paradox in this context. However, productivity gains have not been realized at the same rate in the tropics, at least on average.

Agricultural production, as opposed to productivity (amount produced relative to inputs used), measures total output, whether from intensification or extensification. Figure 4 correlates increases in agricultural production and tropical forest loss. The general relationship is positive; increased production is associated with forest loss. However, there is a wide range of realized deforestation outcomes at all levels of agricultural productivity change, suggesting an underlying mix of intensive and extensive agricultural growth. The section on global context below discusses the role of international trade.

Climate change and a better understanding of ecosystem services have introduced additional dimensions to the valuation of forests, including carbon storage in forests and the impact of forests on rainfall. In the case where these services are priced into land rents, they shift the R^f curve upward. This would reduce F, the point at which the land manager is indifferent between agriculture and forest. Pricing the environmental value of forests leads to greater forest growth in low-productivity agricultural areas, typically associated with extensive technologies.

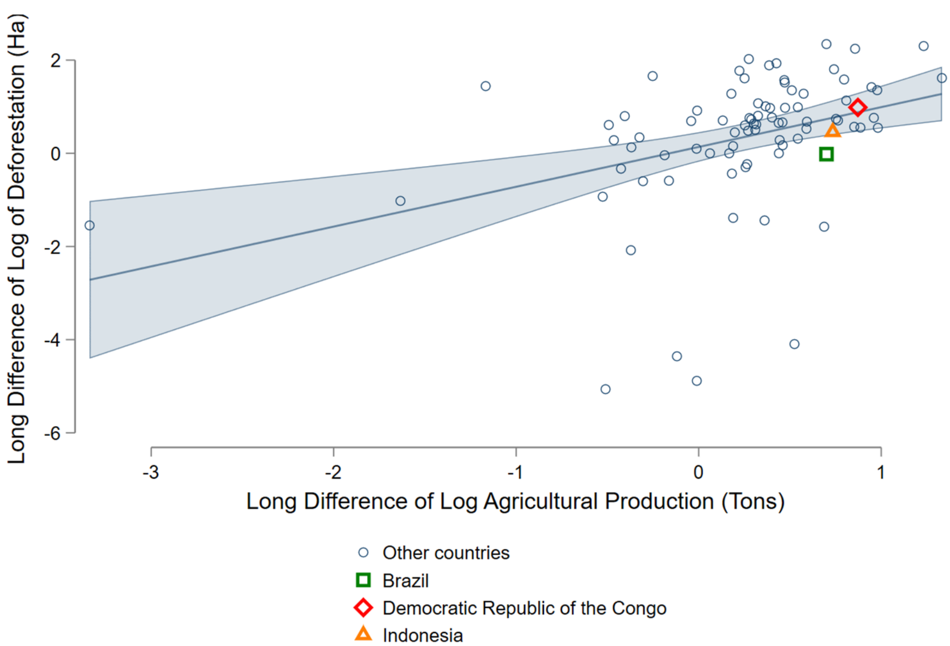

Figure 4 Relationship between long difference in log of deforestation and log of agricultural production for tropical countries (2001–2019). Difference in the log of deforestation (measured in hectares) from 2001 to 2019, calculated as the log of deforestation in 2019 minus the log of deforestation in 2001, and the difference in the log of agricultural production (measured in tons) from 2001 to 2019, calculated as the log of agricultural production in 2019 minus the log of agricultural production in 2001. Deforestation data from Global Forest Watch (2024). Sample: 83 tropical countries. The shaded area represents the 95 percent confidence interval.

The framework is also useful for assessing the impact of policy interventions. Extensions to the model can accommodate credit constraints or other market frictions (Angelsen 2007). In the following sections, we apply this framework to understand how policy tools and changes in the economic context can affect productivity, deforestation, and environmental externalities.

Evidence from Microeconomic Studies

This section presents evidence on the impacts of specific policies intended to increase agricultural production. We include here policies targeting prices, agricultural technology, regularization of property rights, and credit and risk-management instruments, as well as policies directly designed to decrease the environmental damages from agricultural production. For research evaluating policies with multiple goals, we group articles according to their stated primary mechanism.

Output Prices

Prices are a primary lever in determining production choices; in the context of our theory, they shift the level of the agricultural and forest rent functions. Although any policy conducted at scale may affect prices, we focus on output prices in this section. We leave input price subsidies and export price interventions to later sections.

Although agricultural price policy has long been at the center of development policy (Timmer 1986), analysis of this approach has often excluded environmental outcomes. A recent global study links spatially specific crop price indices to global deforestation data, finding that increases in the crop price index explain around one-third of the deforestation observed in the tropics between 2001 and 2018, with stronger associations in areas more open to international trade and in less wealthy places (Berman et al. 2023). Although this article highlights a fundamental agriculture/environment trade-off, crop prices can also change the underlying efficiency of agricultural land use. In Brazil, higher soybean and meat prices and lower wood prices are associated with more deforestation (Hargrave and Kis-Katos 2013; Assunção, Gandour, and Rocha 2015). Further, price decreases can inhibit the profitability of agricultural expansions (Assunção, Gandour, and Rocha 2015).

In addition to price levels, food price stability is strongly associated with food security, political stability, and overall economic growth (Amolegbe et al. 2021). Recent work analyzes the impact of maize price volatility and land-use change across 26 African countries (Lundberg and Abman 2022), demonstrating that price instability leads to less land conversion. These two articles together highlight the tension that can exist between development and environmental outcomes.

Technology Adoption

Much effort has been expended by governments and international agencies to promote appropriate agricultural technology in the tropics. However, empirical evidence suggests that the same policy to increase intensive agriculture can lead to either increased or decreased environmental preservation. This is consistent with the ambiguous theoretical predictions of the land-use effect of increased agricultural productivity. Higher yields or lower production costs may shift the agricultural rent curve to the right, stimulating land-use change, or it may change the trade-offs between intensive and extensive agriculture, resulting in land being spared.

Figure 5 documents increased input use—irrigation and nitrogen fertilizers, respectively—over time. There is a persistent divergence between tropical and nontropical countries. Irrigation coverage has increased everywhere over time, with no sign of the gap closing between tropical and nontropical countries. There does seem to be more convergence in the intensity of fertilizer use, but tropical countries have had a relatively constant rate since the late 1980s. Much of the productivity gap documented in figure 3 is likely driven by differences in input use.

Three challenges have plagued empirical research in this area. First, technological progress in agriculture has stagnated in poor countries, particularly in Africa. Suri and Udry (2022) argue that this is largely because of multiple binding constraints to technology adoption, such as credit constraints, information gaps, and the lack of access to complementary inputs (e.g., training may be a necessary complement to achieve results from an input such as improved seeds). At the same time, measurable productivity improvements over longer historical periods are set against the lack of credible satellite-based measurements of forest cover before 2000 (Potapov et al. 2022). Second, the ideal experimental setting for establishing causal effects is a randomized control trial (RCT), with a comparable treatment group receiving a program (such as training and improved seeds) and a control group continuing as usual. However, many agricultural interventions are not amenable to a randomized design; those that are have had modest productivity effects thus far (Bridle et al. 2020). This makes it difficult to precisely estimate the downstream impacts of an intervention (e.g., deforestation). Finally, measurement error in satellite-based detection of forest loss further results in inconclusive or noisy findings (Alix-García and Millimet 2023).

Different methods have yielded different conclusions about environmental outcomes. For example, Stevenson et al. (2013) simulate a global economic model using national accounts data to argue that the Green Revolution helped avoid 18 to 27 million hectares of deforestation by improving productivity on cultivated land. In an earlier study, however, Foster and Rosenzweig (2003) use instrumental variables regression to show that forest growth in India was lower in areas with larger increases in productivity under the Green Revolution. They argue that increases in Indian forest cover were attributable to increased demand for forest products, spurred by technology-induced income and population growth. Global and regional patterns, however, do suggest that increases in agricultural intensification correspond to reduced deforestation levels (Burney, Davis, and Lobell 2010).

More recently, Szerman et al. (2024) use the expansion of farm electrification to study the effects of increases in agricultural productivity on deforestation in Brazil. They demonstrate that electrification increased farm productivity but not the financial returns to land-extensive cattle production. They find that, in the absence of this increase in productivity, deforestation rates in Brazil between 1985 and 2006 would have been 50 percent larger. A nice feature of this study is that, although it relies on historical improvements in productivity, the authors use satellite imagery to confirm that these effects persisted even 25 years after the productivity shock (i.e., the electrification that increased productivity). Focusing on more recent deforestation in Brazil, Carreira, Costa, and Pessoa (2024) find that Brazilian municipalities that adopted high-yield soy varieties deforested less in response to a large increase in export demand from China.

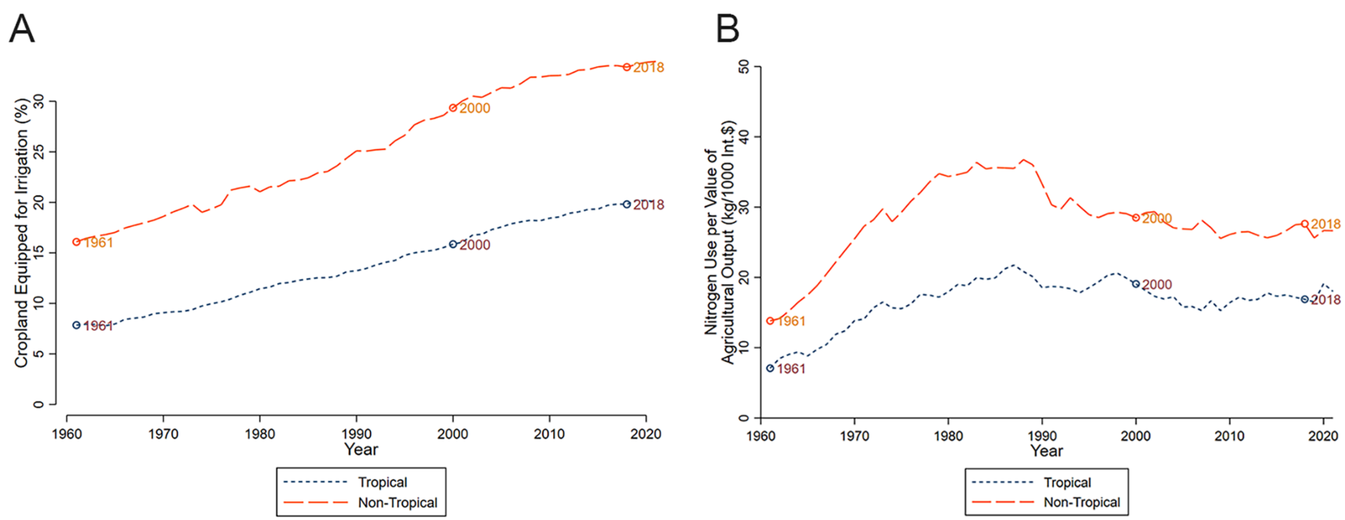

Figure 5 Evolution of cropland equipped for irrigation and use of nitrogen fertilizer. Panel *A* shows the cropland equipped for irrigation (as a percentage of total cropland area) over time. Panel *B* shows the evolution of nitrogen use, measured in kilograms per 1,000 international dollars of agricultural output, where an international dollar is a hypothetical currency representing the purchasing power a US dollar has in the United States at a given time (https://ourworldindata.org/international-dollars). Sample: 115 countries (48 nontropical, 67 tropical) for panel *A* and 105 countries (44 nontropical, 61 tropical) for panel *B*.

Other work leverages variation in input price policy to study the impact of technology on agricultural deforestation. Abman and Carney (2020) use externally determined variation in subsidies for fertilizers and improved seeds in Malawi to show that these productivity-enhancing investments reduced pressure to expand agriculture and thus reduced deforestation. A recent review describes a suite of pre-2000 studies demonstrating a reduction of deforestation in Zambia with the introduction of price and input subsidies, which was then reversed when the subsidies were removed (Holden 2019). Other evidence from Zambia shows a decrease in deforestation associated with farmers' adoption of improved maize seeds, though the opposite correlation exists for fertilizers (Pelletier et al. 2020). Finally, Abman et al. (2020) use a quasi-experimental approach that compares the outcomes of an intervention (a treatment) in one geographic area to the outcomes in a similar place that did not receive the treatment because it was just outside an arbitrary geographical zone established by the program. They show that a Ugandan agricultural training extension program (not to be confused with land extension) reduced deforestation by 14 percent in targeted areas. They are able to detect effects on deforestation in part because the program generated a 32 percent increase in revenue earned by farmers, possibly because it was targeted to low-productivity producers.

Finally, there is intriguing new research on the effects of providing information on the adoption of agricultural water-management techniques that can both raise productivity and combat land degradation. In Niger, an experiment that trained producers in the construction of rainwater collection berms offered training to three groups of women. The first group received only training, the second training plus a cash transfer, and the third training plus a cash transfer conditional on the establishment of berms. The collection of berms resulted in increases in agricultural production and the restoration of previously uncultivable land (Aker and Jack 2025). The information intervention alone increased the construction of berms by 90 percentage points, with no additional effect of cash transfers.

The limited set of articles estimating causal effects of agricultural productivity on deforestation means that there remain open questions. A key gap is identifying whether there are inflection points when the deforestation avoided because intensification is overtaken by increased pressures from extensification.

Transport and Other Infrastructure

The improvement of basic infrastructure fundamentally changes the costs of transport and inputs faced by producers. Recalling our framework, transport costs generate the decreasing rent gradients of the Von Thünen model. The rapid expansion of transportation and other infrastructure in the twenty-first century, coupled with increased availability of spatially explicit digitized data on these investments, has allowed for recent advances in understanding the causal effects of transport infrastructure on agriculture and deforestation, though few studies examine both outcomes. Increases in agricultural productivity and food security as a result of better transport infrastructure are documented by Adamopoulos (2025) in Ethiopia and by Brenton, Portugal-Perez, and Régolo (2014) in Central and Eastern Africa.

The impact of road investments on forest cover is theoretically ambiguous. New roads may lead to forest loss by facilitating market access for timber, firewood, and agricultural expansion or by increasing land value for settlements and industries. Conversely, roads might mitigate forest loss by improving access to substitutes for forest resources and enhancing connections to external markets, which could reduce incentives for land clearing. The effects likely differ depending on whether the roads are rural feeder roads or national highways.

Asher, Garg, and Novosad (2020) examine expansions of two different types of road networks in India: last-mile rural roads connecting villages that previously had only unpaved roads and highway expansions that serve as major transport arteries. Using a population threshold–based regression discontinuity design, they find that rural roads that led to major reallocation of workers from farm to nonfarm employment had zero effect on local forest loss. By contrast, highways dramatically increased deforestation. The result on highways is consistent with previous work in Belize (Chomitz and Gray 2017), Brazil (Bebbington et al. 2018; Pfaff et al. 2018), and Indonesia (Bebbington et al. 2018; Poor et al. 2019). Further, recent evidence suggests that by either ignoring or not fully accounting for general equilibrium effects—the effects on the broader economy—the deforestation consequences of highway construction may be underestimated by up to 25 percent (Araujo, Assunção, and Bragança 2023).

Other work examines specific production infrastructure. Saavedra and Moffette (2026) find that the mass closure of slaughterhouses in Colombia had no effect on deforestation, nor did this decrease herd sizes or total livestock numbers at the municipality level.

In addition to individual types of infrastructure expansion, recent studies examine the effects of programs that encompass many different kinds of infrastructure. Baehr, BenYishay, and Parks (2021) show that irrigation infrastructure reduced forest loss and that rural roads did not increase deforestation, suggesting that tailored policies supporting local infrastructure can generate both economic and ecological returns. In Ghana, Abman and Lundberg (2024) study the effect of guaranteed contracts, pricing, and methods to pick up the output of palm oil producers. Though the analysis does not provide information on increases in oil palm production, it does detect increased forest loss as a result of its expansion.

Property Rights Security

As mentioned, land ownership or tenure rights are not necessarily secure in less developed countries. Farm plots are often allocated through custom rather than formal titling, and some customary rights can be lost if a farmer ceases to work the land for a period of time, perhaps because of an off-farm work opportunity. The risk of losing one's right to a parcel of land discourages investments that might improve productivity. Within a dynamic framework, the risk of future loss of ownership has similar effects to an increase in the discount rate (i.e., a stronger preference for present over future payoffs). Such risk incentivizes the short-run exploitation of a natural resource; for instance, a farmer may be less likely to invest in soil conservation.

However, theory also demonstrates that this effect may be more complicated when considering a broader portfolio of economic activities. Liscow (2013) expands earlier work by Farzin (1984) to highlight the mechanisms through which tenure insecurity affects investments across the land portfolio. In particular, landholders with secure tenure may invest more because (*a*) they can use their land as collateral, (*b*) they feel more secure in making longer-term investments such as improving soil quality, or (*c*) they do not have to waste energy defending their property. Whether this leads to a larger area in agriculture or in forest depends upon the returns to investment in each sector. Within our framework, one way to think about this is that the expected rent curves can be shifted outward by secure property rights (reflecting higher rent for all activities), and the final effect on forest is determined by the relative magnitude of these shifts. When the returns to investment in agriculture are higher than those in the forest sector, it may be the case that stronger property rights lead to more agriculture at the expense of forests. Liscow (2013) provides evidence of this dynamic in Nicaragua. Tseng et al. (2021) provide an excellent review of studies published between 1990 and 2018 that examine the relationship between land tenure security and environmental and human outcomes. Broadly, they find that improvements in property rights improve environmental outcomes and human well-being, including increases in agricultural incomes. They find no cases where an improvement in land tenure security decreased agricultural productivity and a preponderance of situations in which environmental outcomes improved as well. This review also highlights the scarcity of studies that examine agricultural and environmental outcomes jointly using rigorous experimental design.

Among the few property-rights interventions that examine both agricultural productivity and environmental outcomes is an RCT in Benin that delineated the boundaries of around 70,000 landholdings. Deforestation was significantly lower in villages randomized into the program (Wren-Lewis, Becerra-Valbuena, and Houngbedji 2020). Complementary household survey data indicated that households were less likely to clear land as a means of securing it and more likely to regulate the community's forest access to conserve forest resources; they also reported higher levels of trust. A separate study of the same program demonstrated higher investment in agriculture but no short-term increase in agricultural profits (Goldstein et al. 2018). An RCT in Zambia reveals a different type of effect: increasing land tenure security via demarcation of property lines increases recipients' sense of security but has no impact on long-term investments, including in agroforestry (the integration of forestry and farming; Huntington and Shenoy 2021). Finally, Lipscomb and Prabakaran (2020) use a quasi-experimental approach to examine the rollout of a land registration program in the Amazon. On average, there is no deforestation effect, but they do show a decrease in the area under temporary crops. Counties with higher numbers of registrants show decreases in the deforestation rate, consistent with a setting in which agricultural production is used to secure tenure.

In sum, the variety in findings likely reflects contextual differences across studies. It would be useful to have more causal evidence characterizing the specific conditions that have led to tenure security improving or worsening agricultural and environmental outcomes.

Agri-Environmental Interventions in the Tropics

In the United States, Canada, and Europe, policies to limit the negative externalities from agriculture and also support rural households have become increasingly common (Baylis et al. 2022). These include crop-diversification requirements within the European Common Agricultural Policy, as well as the US Conservation Reserve Program, which restores natural vegetation on degraded land. A key question is whether such policies can simultaneously support farmer incomes and conserve environmental services. Much existing evidence is based on case studies or correlational collections of projects rather than on rigorous causal analysis (Pretty et al. 2018).

In the tropics, interventions promoting land-sparing agricultural intensification include technological and conservation solutions. In Brazil, an RCT evaluating a large-scale training and technical assistance program found that affected farmers simultaneously increased revenues and carbon sequestration. Farmers were required to learn about one of four sustainable land-management practices, and a subset were offered visits by field technicians to customize their advice. A key finding was that training alone did not have any impact, but 24 monthly extension visits generated substantial treatment effects on pasture restoration, rotational grazing, improved management and conservation practices, more tractor use, and expenditures (Bragança et al. 2022). These results contrast with a payment program intended to encourage sustainable cattle production in Mexico, where cash transfers were allocated to current livestock producers but were not accompanied by training or technical assistance. In this case, there is no evidence that intensification ensued; in fact, municipality-level deforestation increased as a result of the program (Moffette and Alix-García 2024).

Informational interventions have had some successes in increasing the input use efficiency. Such interventions aim to increase or maintain yields and reduce excessive input use linked with environmental externalities. In Nigeria, Arouna et al. (2021) report that an RCT randomizing personalized fertilizer advice via a mobile app increased rice yields and profits without increasing overall fertilizer use. In Bangladesh, implementation of a simple leaf color chart to guide urea application reduced fertilizer use without lowering yields (Islam and Beg 2021).

Burning agricultural residue prior to planting, a common practice across the tropics, generates externalities through carbon emissions (Cassou 2018), cognitive performance (Zivin et al. 2020), and health (Behrer 2019). One intervention has been payments for ecosystem services (PES), in which land managers are offered payments to achieve environmental goals. There have been mixed results in recent RCTs that attempt to refine PES-type contracts for farmers to reduce burning. A project in Indonesia that offered training, grants for firefighting, and the promise of a payment conditional on *no fire* at the end of the growing season generated no detectable effects on fire outcomes (Edwards et al. 2024). However, an RCT offering up-front versus ex post payments to reduce burning in India resulted in high take-up of partial up-front payments with no commensurate decrease in agricultural yields. Although ex post payments had no effect on burning, partial up-front payments reduced the incidence of burning by around 50 percent (Jack et al. 2022).

Agroforestry, an agricultural system integrated with trees, is widely promoted in the tropics. There is significant research on the impacts of agroforestry on outcomes such as soil fertility and crop yields but scant assessment using robust causal methods to measure its effects on both agricultural income and environmental factors. One exception is Hughes et al. (2020), who evaluate an agroforestry extension program in western Kenya. The program trained farmers in agroforestry management and incentivized tree planting with modest payments upon confirmation of tree planting. The evaluation uses a quasi-experimental design to uncover causal impacts after 10 years. The estimated effects on agricultural outcomes were positive; they included increases in the value of fuelwood and decreases in time spent collecting it, increased use of fodder by dairy farmers, and higher milk yields. Still, there is a significant evidence gap in this literature. A recent systematic review located only eight quasi-experimental studies of agroforestry interventions and no RCTs (Miller et al. 2020). Only two of these measured environmental outcomes are beyond the adoption of the agroforestry practice.

In an attempt to encourage intensification of soy and cattle production in Brazil and simultaneously limit deforestation, industry leaders signed commitments to produce only on land that had been deforested prior to the agreements. Evidence documents the displacement of soy production in the regulated area onto pastures in that same region, pushing cattle production into an unrelated zone and inducing deforestation (Moffette and Gibbs 2021). Two other articles find that environmental policies can increase agricultural productivity by making deforestation more costly via punishments for cutting down trees or payments for conserving them (Koch et al. 2019). This may induce increased use of agricultural credit and therefore investment (Moffette, Skidmore, and Gibbs 2021).

In sum, existing work suggests that interventions that convey knowledge, offer continued technical support, and provide financial incentives may increase productivity and preserve ecosystem services. Eliminating one of these elements may compromise success. There remains substantial room for causal evidence on the environmental and agricultural productivity effects of agroforestry and for studies on PES-type programs that also examine food production outcomes.

Credit and Risk-Management Instruments

Proper access to financial services has long been recognized as a critical factor in the development process, particularly in addressing credit constraints and managing risks (Townsend 1995). Improved credit and risk management increase returns on agricultural investments and can shift the intensive agricultural rent curve upward (increasing rents, all else equal), thus reducing the pressure for clearing forest. Greater availability of insurance can also decrease the use of forest as an income source of last resort. For farmers deciding between intensive and extensive agricultural practices, the ability to cover up-front costs or bear higher risks can be crucial in enabling sustainable increases in agricultural production. For example, comparing equilibrium models of the United States and India, Donovan (2021) suggests that incomplete credit markets decrease agricultural productivity by 16 percent, primarily by reducing the usage of intermediate inputs (and thus intensive production technologies). In influential work in Ghana, Karlan et al. (2014) demonstrate that uninsured risk is a major barrier to investment, possibly more crucial than improving access to credit in that particular context. Achieving higher food production and preserving ecosystems requires greater capital investment per hectare and the adoption of practices that may involve higher exposure to natural risks.

The evidence on the direct relationship between expanded credit access and deforestation shows mixed results. Exploiting a 2008 policy that conditioned subsidized credit in the Amazon to farmers with legal paperwork and environmental conformity, Assunção et al. (2020) used a differences-in-differences approach (comparing changes over time between a quasi-experimental treatment and control group) to show that municipalities with less credit reduced deforestation. On the other hand, Assunção et al. (2019) use the combination of aggregate fluctuations across banks' funding and local configuration of bank branches serving each municipality in Brazil to predict credit expansion. This method uncovers causal effects by eliminating bias that comes from the fact that credit may expand more quickly in places with higher agricultural productivity. Municipalities with better access to credit expanded production through yield gains, reducing the pressure on deforestation. This result holds largely for areas outside the Amazon, where historical land use has left extensive unproductive cleared areas, creating opportunities to boost agricultural production through productivity gains. In the Amazon, Ruggiero, Pereda, and Pfaff (2025) document heterogeneity in the effects of credit expansion on deforestation. Long-settled areas show no environmental effects of credit expansion, whereas areas with existing crop production suffered more deforestation as agricultural credit spread; in more remote areas, credit for cattle has led to increases in deforestation.

The evidence directly linking financial market improvements to agricultural and environmental outcomes is weak. However, taken as a whole, this evidence reveals that policies that relax the financial constraints that are pervasive in developing countries should take into account possible threats to forests and other ecosystem services.

Global Context

Most of the evidence we have reviewed so far focuses on micro-elements of the agricultural production decision. However, these decisions take place within a global context where comparative advantage (a country's production efficiency compared with other countries in a particular economic sector) and trade barriers are primary forces in determining local food prices. The tropics are home to both large amounts of agriculture (especially small-scale subsistence agriculture) and critical environmental services through unique tropical biomes. By virtue of the fact that tropical forests exist only in the tropics, the region has a comparative advantage in tropical forest production and related ecosystem benefits. Thus, for this land to be deforested for agriculture, agricultural production must also be cheaper in these regions relative to places that produce the same crops. Puzzlingly, agriculture as a share of employment is high in the tropics, yet agriculture is lower in relative productivity than other sectors in the same countries. According to Lagakos and Waugh (2013), adjusting for prices, the gap in aggregate output per worker between the 90th (highest) to 10th percentile (lowest) of the world's income distribution is 45 to 1 in agriculture but just 4 to 1 in nonagriculture. Yet agriculture's share of employment averages 65 percent in 10th-percentile countries and only 3 percent in 90th-percentile countries.

Given this evidence, standard Ricardian economic models would suggest that workers should leave agriculture (and thus preserve more forest). Some friction or unobservable factors thus prevent reallocation in line with economic intuition. The next sections link these macro-level agricultural productivity disparities with environmental outcomes in the tropics.

Trade

Understanding the link between trade and food production is fundamental to the environmental implications of tropical agriculture. Resource extraction in the tropics can be efficiently averted only insofar as there is cleaner production elsewhere. Brander and Taylor (1995) put forward a fundamental framework linking comparative advantage and resource extraction. The authors emphasize two predictions regarding the effect of trade on environmental degradation. First, if the productivity losses from deforestation are not severe and regulation is quite costly, then there will be open access to forest. Second, if deforestation had taken place in the tropics prior to trade liberalization, but the Global North is agriculturally more productive than the Global South, then liberalization would be good for environmental outcomes.

Recent work by Farrokhi et al. (2024) tests the Brander and Taylor (1995) framework in a modern global computable general equilibrium model (a simulation of economic impacts). They focus on the effects of liberalizing trade between Brazil and the European Union: a 30 percent reduction in costs in global agricultural trade results in an increase in steady-state worldwide forest share by 0.8 percentage points. A key insight of their analysis is that trade liberalization should be deforestation-reducing when absolute advantage and comparative advantage in agriculture align (when a country is both very efficient in agricultural production and, compared with other countries, gives up less production in other sectors to add a unit of agricultural output). However, if absolute advantage in agriculture is misaligned with comparative advantage, trade liberalization will push deforestation across borders to regions of high comparative advantage. The authors do not directly calculate the food costs of deforestation induced by trade policy.

Two separate articles calculate how food production is affected by trade but do not address deforestation. Tombe (2015) argues that agricultural trade costs tend to be higher in countries with a higher subsistence share of agriculture. Given that subsistence agriculture tends to use extensive practices, this would suggest that trade policy is intensifying deforestation. Nath (2025) argues that climate change will only exacerbate this problem unless trade barriers are relaxed. More systematic evidence regarding a joint food–deforestation trade-off is needed to tie together these results and those of Farrokhi et al. (2024).

In response to growing discussions regarding trade policy as a deforestation remedy, Harstad (2024) proposes dynamic conservation contracts to protect forest. He argues that static PES contracts can be expensive and hampered by shifting political priorities, as they require repeatedly paying a targeted country for environmental services, whereas the targeted country has a one-shot deforestation decision at any point. He proposes a conditional trade agreement that provides favorable terms of trade for an extracting country in perpetuity, contingent on their keeping resource extraction below some threshold. Theoretically, between two countries, this contract achieves the first-best outcome. However, in more general environments, deforestation can "leak" or shift to other countries.

Standard trade agreements favoring the production of agricultural goods appear to fail to balance out environmental concerns. Using a panel of 189 countries from 2011 to 2012 and the enactment of regional trade agreements (RTAs) as a quasi-experiment, Abman and Lundberg (2019) find that agricultural land conversion increased by 0.8 percentage points over the 3 years following the RTAs. Interestingly, this effect is driven by developing tropical countries and almost fully by agricultural expansion rather than by an increase in forest market expansion. A follow-up study revealed that RTAs with environmental provisions effectively mitigated forest loss from trade liberalization, seemingly by limiting agricultural extensification (Abman, Lundberg, and Ruta 2024).

Although this result suggests that deforestation might be, in turn, reversed by punitive tariffs, Hsiao (2026) argues that such corrective trade policy needs both to be coordinated and to have a dynamic commitment mechanism. In the setting of Indonesian palm-oil-driven deforestation, he combines a dynamic model of palm-oil supply and a model of oilseed demand with rich cross-oil substitution characteristics to project the deforestation impacts of punitive tariffs. Unilateral trade policy by the European Union alone achieves only approximately 20 percent of the gains of a fully internationally coordinated tariff schedule. These insights connect the empirical evidence from Abman and Lundberg (2019) with an example of dynamic commitment from Harstad (2024).

This global view suggests that price or trade policy can be effective in combating tropical environmental externalities without sacrificing food production. A first-best solution would be a carbon tax directly on food producers, ensuring that producers face land prices that directly reflect the environmental costs of agriculture. However, recent work has shown that distortions in the existing food supply chain can undercut the effectiveness of such policy. In South America, for example, Dominguez-Iino (2024) compares this first-best solution with a tax on intermediaries in the supply chain who also have market power (the ability to charge prices above the marginal cost of producing goods because of a lack of competition). He argues that taxes on downstream producers (e.g., beef processors) are politically more expedient than taxes on upstream producers (e.g., cattle ranchers) but that these downstream taxes risk leakage—the movement of environmental harm outside the regulated context—because of market power. This happens as follows. Because downstream producers have market power, they have a larger margin between their output price and their break-even cost. Consequently, profit-maximizing producers reduce the pass-through of the carbon tax to upstream ranchers; therefore, ranchers no longer face the full environmental cost of their deforestation, relative to the first-best policy. The author estimates that, in the South American context, the tax signal is reduced by half as a result of this distortion.

Agricultural Price Policy

Agricultural price policy is vastly different across the tropics and nontropical countries. Adamopoulos and Restuccia (2014) measure policy-induced distortions as the cost-adjusted difference between farm-gate and international prices. The proposed distortions explain half of the differences in average farm size and agricultural productivity across rich and poor countries. Extensive barriers to large farming in developing countries, such as limitations on farm size and progressive taxes on agricultural land (intended to reduce the burden on poorer but less productive farmers), misallocate agricultural resources. New evidence links this potential policy-induced misallocation to explicit environmental outcomes. Carleton, Crews, and Nath (2024) demonstrate that agricultural subsidies have tended to concentrate production in groundwater-depleting regions, like India's Punjab, despite the potential for more efficient cultivation. In a global context, efficient reallocation of agricultural cultivation is hampered by these distortions to agricultural incentives.

In summary, the global context of tropical food production is a growing field, with research in both macroeconomics and trade as valuable complements to the evidence presented in prior sections. We highlight opportunities to link the microeconometrically identified phenomena that we discuss in prior sections to the large-scale, general equilibrium comparative advantage framework summarized in this section.

Discussion and Future Research

The question of how to grow enough food and limit damage to the ecological system that supports human well-being is foundational to sustainable development. Our review finds very promising work in this area and also highlights the need for increased investment in research that examines agricultural and environmental outcomes simultaneously.

When we divide the studies that we examine by theme and location, some stylized facts emerge. In general, there is relatively more evidence on technology, agri-environmental interventions, and international trade, although in this last category, many of the studies are simulations rather than direct empirical evidence. Latin American countries are well-represented across categories, and there is substantial evidence from African countries on prices, property rights, technology, and infrastructure. Studies taking place in Asia are under-represented in most categories (figure 6).

Infrastructure improvements, input subsidies, the formalization of property rights, and agricultural extension efforts supporting more sustainable livestock management can yield improvements along both dimensions in some situations. However, it is also clear that the impact of these interventions varies according to context; for example, sustainable livestock-management interventions have been successful in improving productivity outcomes without compromising forest cover in Brazil but not in Mexico. One reason for this seems to be sustained education on the implementation of practices in the former case but not the latter. Similarly, the evidence on road expansion is mixed. This suggests a need to design and study policies within the context of specific countries and their particular market failures, with an emphasis on understanding the specific features that lead to success in a given setting. A deeper

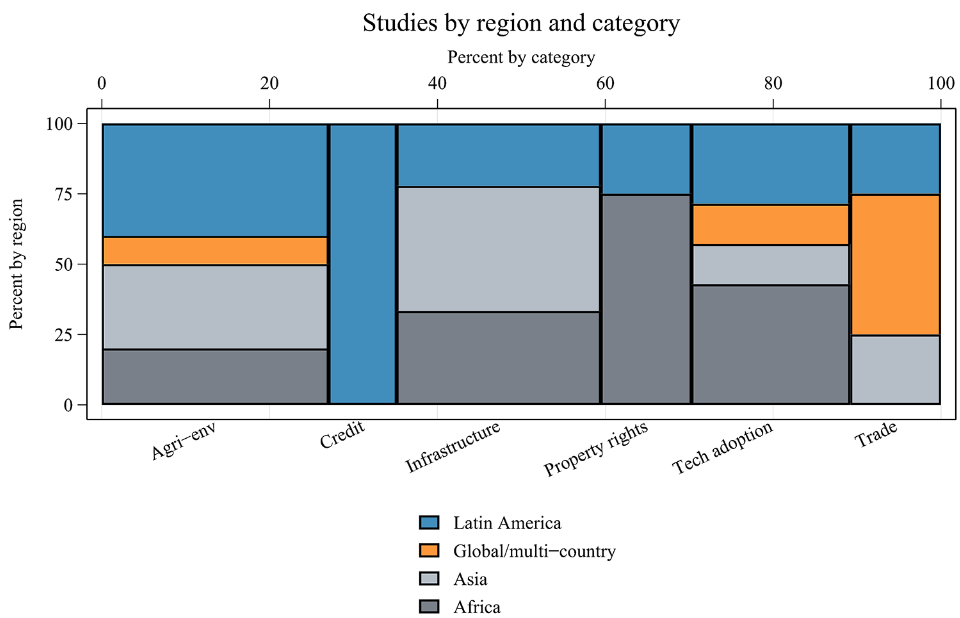

Figure 6 Studies by theme and geographic location. Aggregation of the studies cited in table A2 (tables A1 and A2 are available online) by theme and region. Studies that cover multiple countries across different regions are categorized as global, whereas those that examine countries within the same region are categorized within that region. We exclude review articles from this figure and count only articles or article pairs that consider both agricultural and environmental outcomes.

analysis of mechanisms may help yield generalizable principles of design that can support win-win policies.

The evidence is weak on technologies to increase agricultural productivity. There is significant research in this area that we do not review because it does not include proxies for environmental outcomes. This omission is in itself revealing. Beyond disciplinary disconnects, some of this is because of the nature of RCTs. These tend to be relatively small-scale, with small impacts. They often do not contain spatially explicit information on plot location. In addition, many environmental effects are longer-term, have spatial spillovers, and require large samples to detect impacts. All of these features mean that it is difficult to produce work that examines the environmental effects of agricultural productivity increases. It is crucial to expand this body of work by incentivizing localized measurement of environmental outcomes.

Agri-environmental policies are a promising avenue for joint success in tropical countries. The evidence suggests that careful design, including a combination of technical assistance and financial incentives may be fruitful. Informational interventions have demonstrated positive yield and environmental effects associated with more efficient fertilizer use. However, there is ample room for further evidence, both geographically and in terms of design elements.

Efforts to achieve the twin goals of food and environmental quality cannot escape their broader context. Global analyses highlight the role of distortions in getting food to where it is most needed, and they predict that the tropics should not use as much land for agriculture as they currently do. Theory highlights an important role for trade policy, although empirical evidence suggests that liberalizing agricultural trade requires coordinated and targeted policy to successfully avoid adverse environmental impacts. There is substantial room for evidence on how technology and market power shift food production across space; we know that both of these elements have created distortions, and the question remains how to design policy to reduce these. Finally, global evaluations of misallocation in agriculture have developed rich insights regarding the role of land and agricultural subsidies, but the environmental impact of these large-scale policies in full, global general equilibrium is still an open question. As indicated by the micro-evidence, effectively targeting these existing policy levers can generate win-win policies for agricultural yields and environmental outcomes, but more evidence is needed on how to scale such interventions.

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